

Exception Handling and Modules|Lecture

- Modules in Python
- Generation of random numbers
- Exception handling
- try - except
- Raising Custom Exceptions
- recursive functions



① Modules :- $\hookrightarrow$ a file containing Python code

$\swarrow \downarrow \searrow$
for class var

$\hookrightarrow$.py extension

Purpose $\hookrightarrow$ to organize the code into
sensible and manageable pieces

Types of Modules

$\nwarrow$ math, random, datetime $\leftarrow$ ① Built in modules

$\nwarrow$ ② user defined modules

* ③ Third party modules $\nwarrow$

numpy, pandas, flask

Pip install THAT

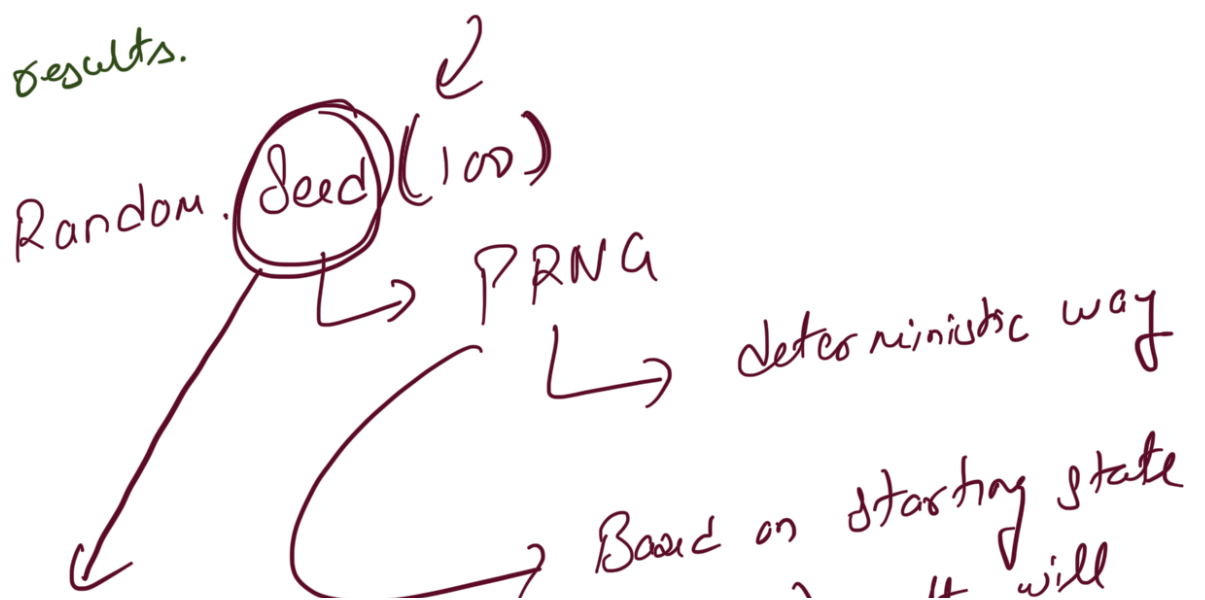
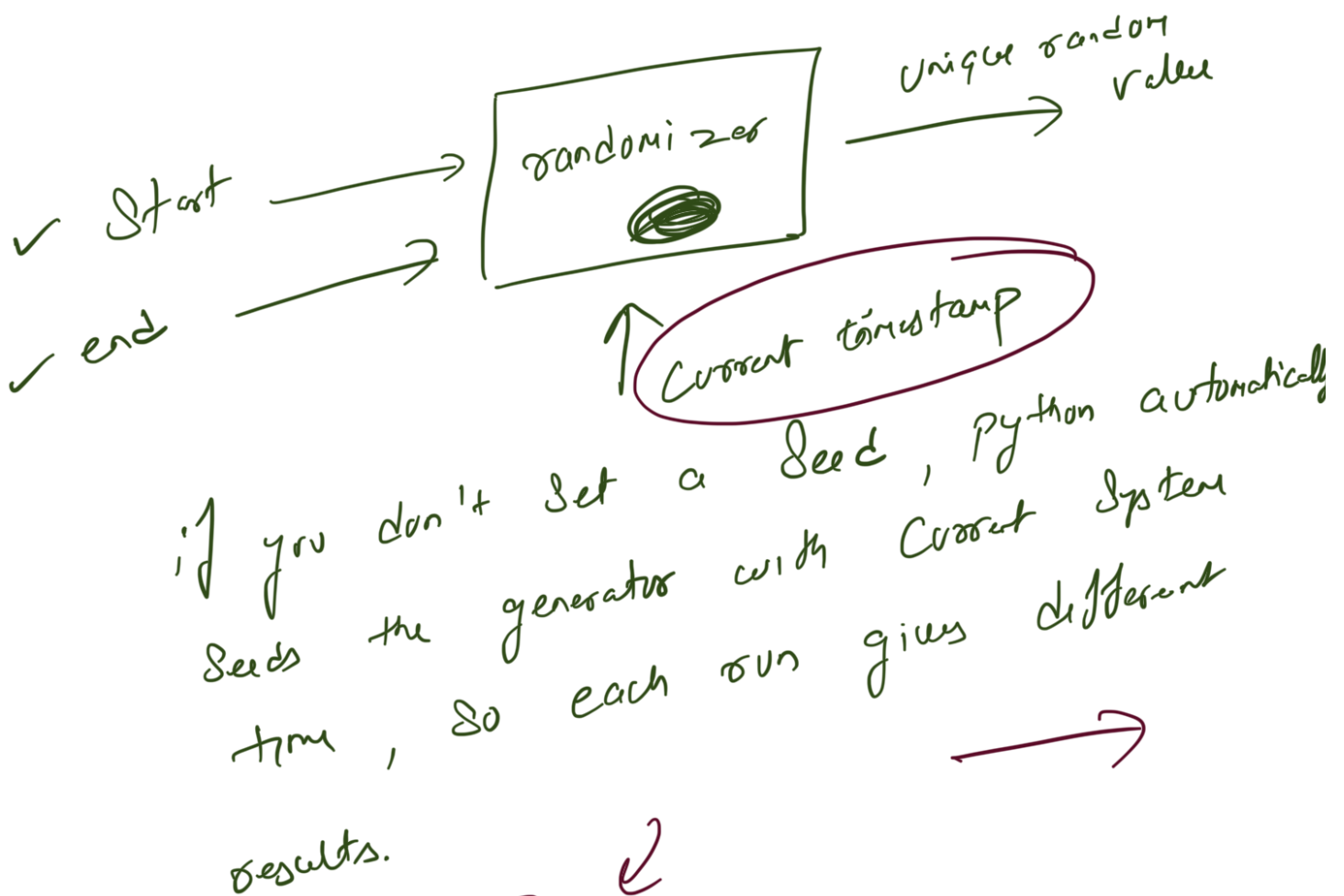
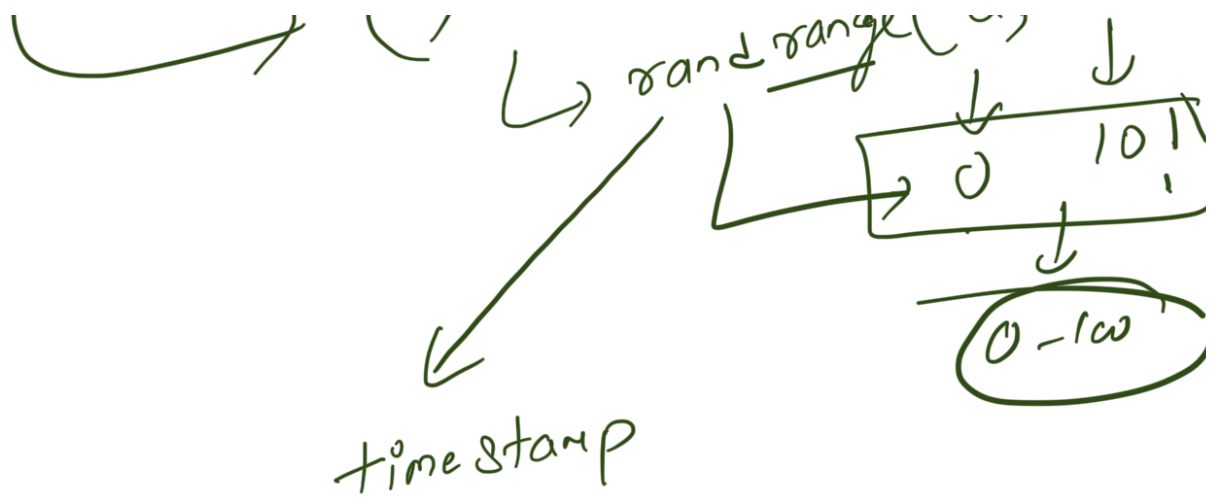
$\hookrightarrow$ import random
Random.random() (0, 100)

$\hookrightarrow$ loads rand module

$\hookrightarrow$ PRNGs $\rightarrow$ pseudo-random
number generators

(0, 100)

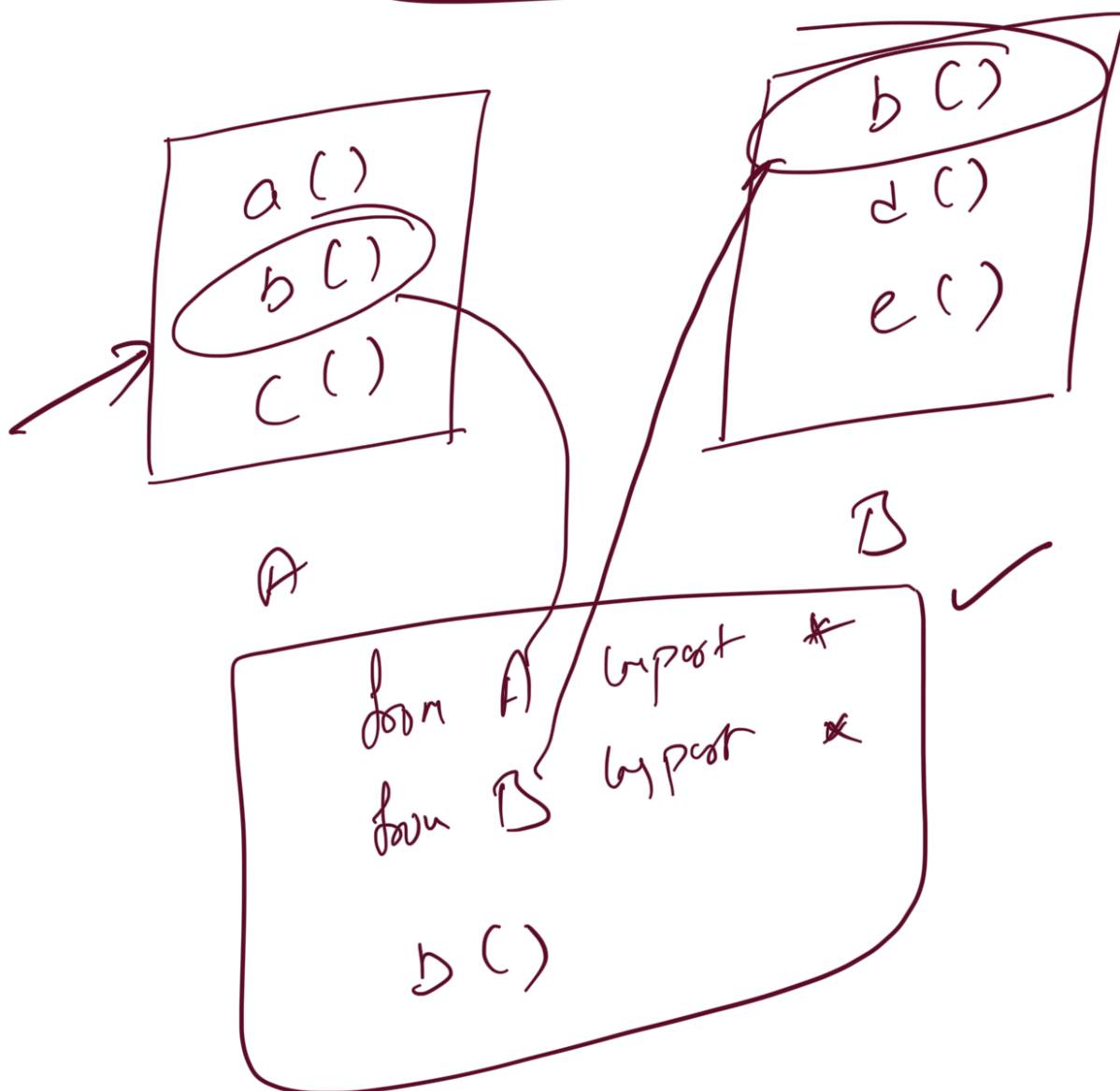
(a, b+1)



Seed is telling
python where to
start with

(seed), ^{no}
always produce the same
sequence of random numbers.

if you use same seed,
python will start from
same position,
you get same sequence of nos.



Exception Handling

Error	Exception
→ Error means something went wrong in program → Most errors are actually represented as exceptions. → Syntax errors: code structure - Run-time errors: mistakes with program runs	→ An exception is a specific object that python creates whenever an error happens at runtime. → Exception can be caught and handled gracefully

Try - except

if exception
do this

else

continue:

Finally: always runs, no matter what happens



try
except
finally

try block can have:-

→ except → to catch errors
→ else → runs only if
no error

→ finally

Finally block helps developer to cleanup
tasks

↳ closing file

↳ disconnect connections with DBs.

↳ Releasing resources.

Raising Custom Exceptions

Built in Exceptions:- not always
describe business
specific problems

Create our own Custom classes